Final Project Report

Time-series data and application to stock markets

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INTRODUCTION

This report is part of a deep learning assignment that explores how deep learning can be applied to stock market analysis. It includes several sub-tasks, each designed to test different methods. The report focuses on explaining the results, discussing any failures, and sharing key observations from each task. The goal is to understand how deep learning can help predict stock market trends and what challenges come up along the way.

TASK 1

Task 1.1

The primary objective of Task 1.1 is to modify the provided demo code to enhance the Nasdaq stock price prediction model by utilizing multiple features. These features include Low, High, Open, Close, Adjusted Close prices, and Volume, rather than relying solely on one feature, such as the Open price. For this task, we used a historical stock dataset of Apple (AAPL) to train and evaluate the model.

The dataset was divided into three distinct sets: training, validation, and test sets. The training set is used to fit the model, the validation set is used for hyperparameter tuning and to prevent overfitting, and the test set is reserved for evaluating the model's generalization performance. The visualization in Figure 1 illustrates this division, where:

- The training set is shown in red,
- The validation set is represented in blue, and
- The test set is marked in green.

The clear segmentation ensures the integrity of the modeling process and enables an accurate assessment of the model's performance on unseen data.

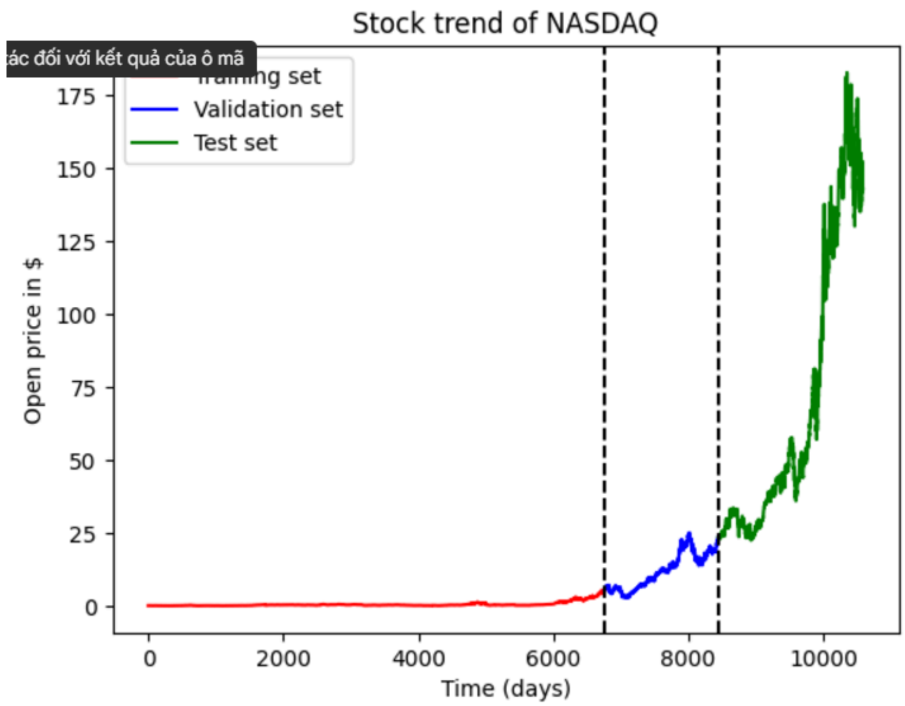

Figure 1. The Stock Trend of the NASDAQ dataset in the training set, validation set, and test set.

During the training phase, we evaluated the performance of two deep learning architectures, LSTM (Long Short-Term Memory) and Conv1D (1D Convolutional Neural Network), to determine which model was better suited for stock price prediction. The results of both models were visualized in Figure 2, where:

- The orange line represents the Real Price,
- The blue line corresponds to the Conv1D Predicted Price, and
- The green line shows the LSTM Predicted Price.

The visualization shows that the Conv1D architecture closely aligns with the Real Price compared to the LSTM model. The Conv1D model successfully captures the temporal dependencies and trends in stock prices, while the LSTM model shows significantly lower predictive accuracy, as demonstrated by the deviation in its predictions. Moreover, the MSE of the Conv1D model is much smaller than LSTM, which contributes to the best performance in stock prediction:

- Conv1D Model MSE on Test Set: 28.202942241745347
- LSTM Model MSE on Test Set: 5264.981028113921

Based on this comparison, Conv1D was chosen as the preferred architecture for training the stock price prediction model in Task 1. This choice ensures better performance and more reliable predictions, as confirmed by its alignment with the actual price trends.

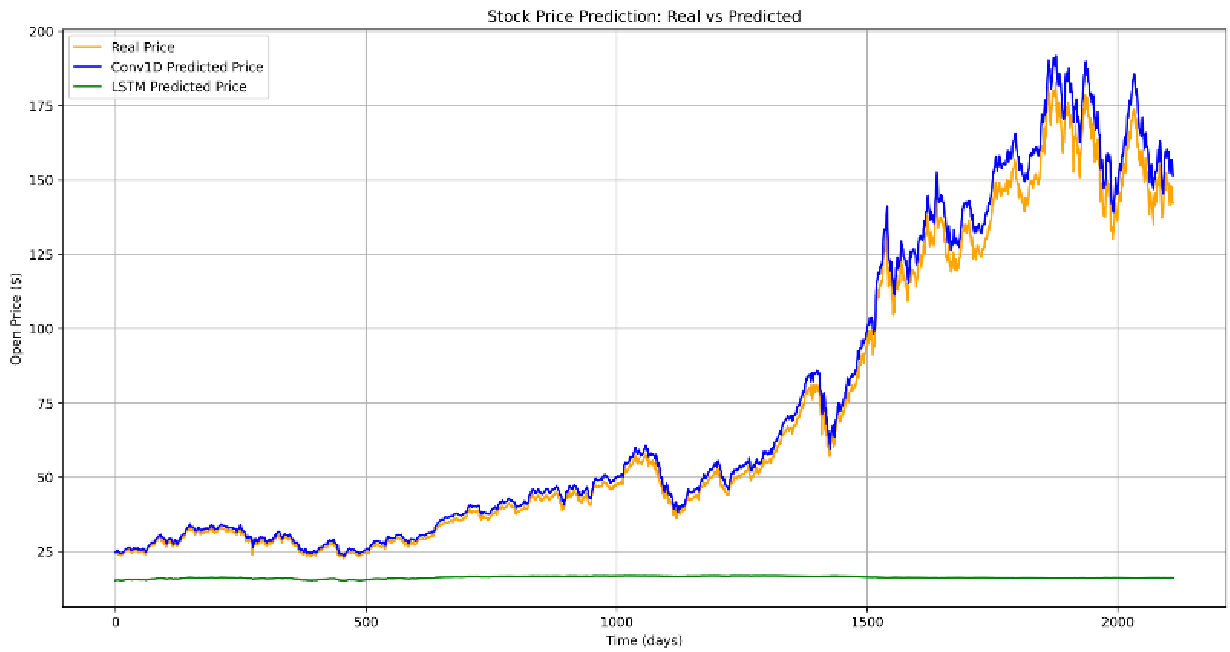

Figure 2: The Comparison of Stock Prediction by LSTM and Conv1D.

Task 1.2

In Task 1.2, the objective was to extend the stock price prediction model to forecast the future Open price of a kth day. For this task, we continued to use the Conv1D model architecture, which had previously demonstrated strong performance in Task 1.1. The model was trained on sequences of 30 days of historical data to predict the Open price three days and seven days ahead.

a) When k = 3

Figure 3 visualizes the model's performance on the test set:

- The orange line represents the Real Price (ground truth),
- The blue line represents the Predicted Price for three days ahead.

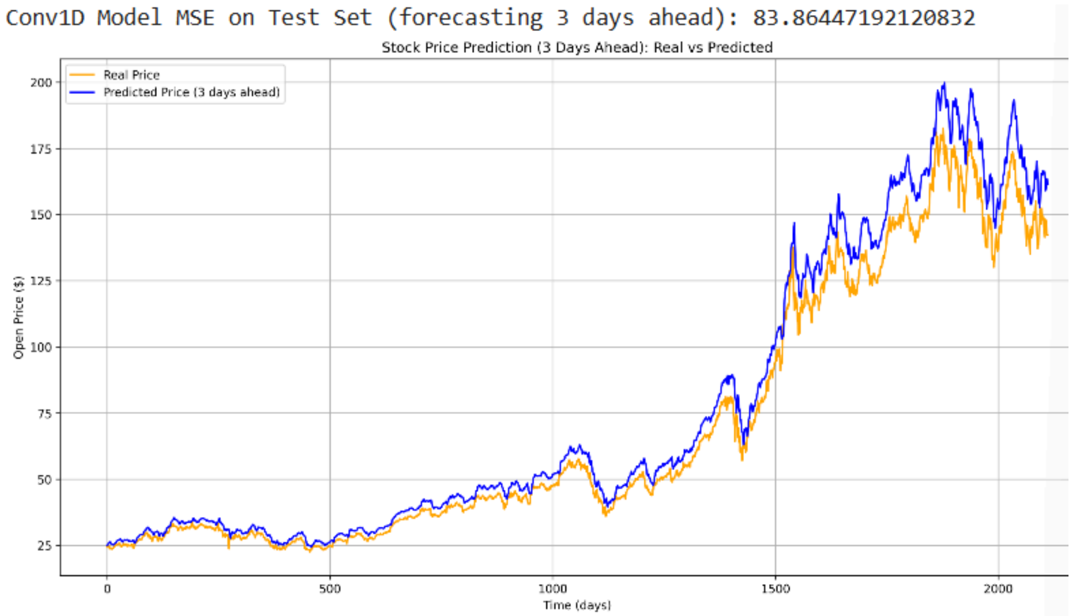

Figure 3a: The stock prediction on the 3rd day when batch size is 250.

From the figure, it is clear that the Conv1D model successfully captures the general trends in stock price movements for short-term forecasting, as evidenced by the close alignment between predicted and real prices. The Mean Squared Error (MSE) on the test set, as indicated in the title, is 83.8645, which reflects the model's ability to generalize to unseen data. This result demonstrates the effectiveness of using the Conv1D model for multi-step time-series forecasting, particularly for short-term horizons like three days ahead.

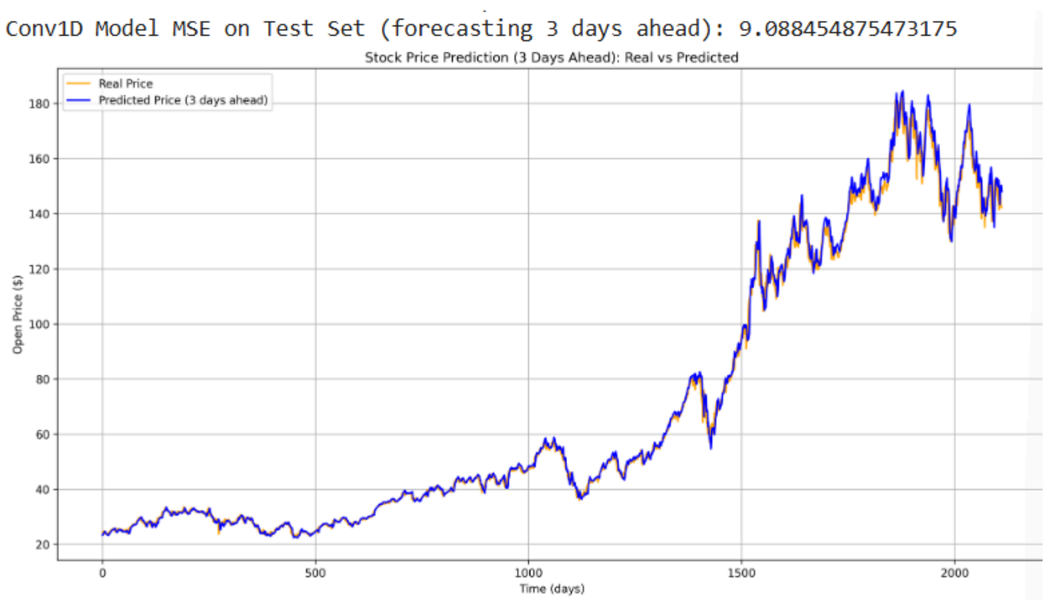

Figure 3b: The stock prediction on the 3rd day when batch size is 64.

Figure 3b shows the performance of the same Conv1D model but with a reduced batch size of 64. The alignment between the predicted prices (blue line) and the actual prices (yellow line) has improved significantly, as evidenced by a much lower Mean Squared Error (MSE) of 9.0885. This demonstrates that reducing the batch size allowed the model to learn more detailed patterns in the data, leading to better overall performance. The closer match between the two lines in Figure 3b reflects enhanced accuracy, which emphasizes the importance of tuning hyperparameters like batch size for improved forecasting results.

By comparing these two results, it is evident that optimizing the batch size significantly impacts the model's accuracy and predictive performance by reducing MSE and improving alignment with real data trends.

b) When k = 7

Using the Conv1D architecture, the model was trained on sequences of 30 days of historical data to predict the stock's Open price after 7 days, implementing a medium-term forecasting approach.

Figure 4a showcases the model's performance on the test set:

- The orange line represents the Real Price (ground truth).
- The blue line represents the Predicted Price for seven days ahead.

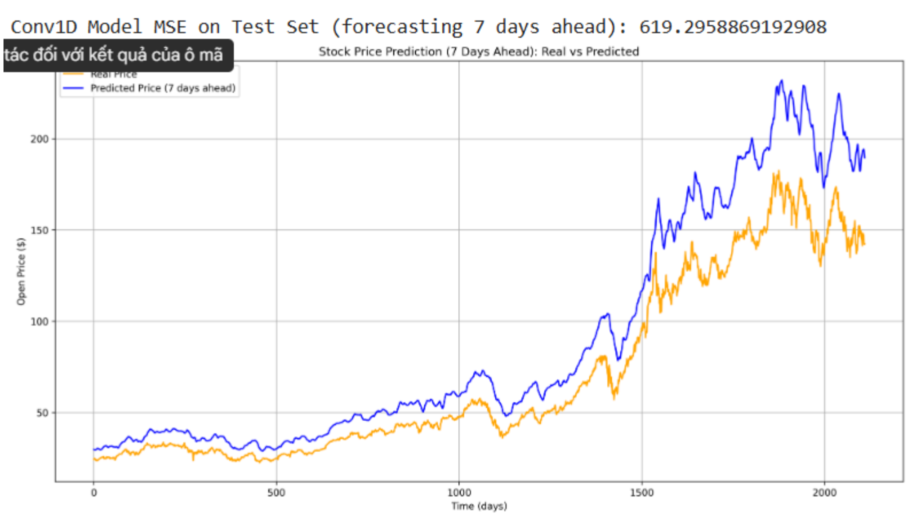

Figure 4a: The stock prediction on the 7th day when batch size is 250.

From the visualization, it is evident that the model captures the general trends in stock price movements but struggles with noticeable deviations in highly volatile periods. This reflects the inherent challenges of predicting stock prices over a longer horizon due to the increased uncertainty and the compounding of prediction errors. The Mean Squared Error (MSE) for this

7-day forecast was 619.296, which is significantly higher than the MSE observed for shorter-term predictions like the 3-day forecast. This result aligns with the expected difficulty of forecasting prices further into the future.

When the batch size was adjusted from 250 to 64, the model's performance improved significantly, as shown in Figure 4b.

Figure 4b illustrates the improved 7-day forecast:

- The MSE reduced to 34.783, reflecting a substantial decrease in error compared to the larger batch size.
- The alignment between the Real Price (orange line) and Predicted Price (blue line) is noticeably closer, suggesting better generalization and accuracy with the smaller batch size.

These findings emphasize the sensitivity of the Conv1D model to hyperparameter tuning, particularly batch size, in achieving improved performance for medium-term stock price forecasting.

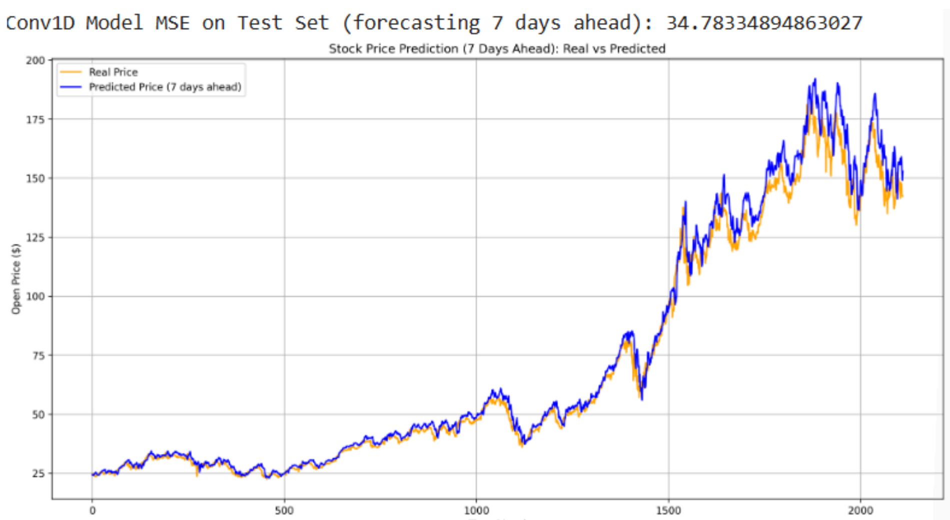

Figure 4b: The stock prediction on the 7th day when batch size is 64.

Task 1.3

a) When k = 3

Using the Conv1D model architecture, the task involves forecasting the stock price for three consecutive days (k = 3) based on historical data. Figure 5a displays the results, where:

- Dashed Lines represent the Real Prices for the 1st, 2nd, and 3rd days ahead.
- Solid Lines represent the Predicted Prices for the corresponding days.

The model demonstrates a high degree of accuracy, with the predicted lines closely following the real prices over the three-day horizon. The Mean Squared Error (MSE) on the test set is 10.099, reflecting the effectiveness of the model in capturing short-term trends. This low MSE highlights the robustness of the model when using a batch size of 250.

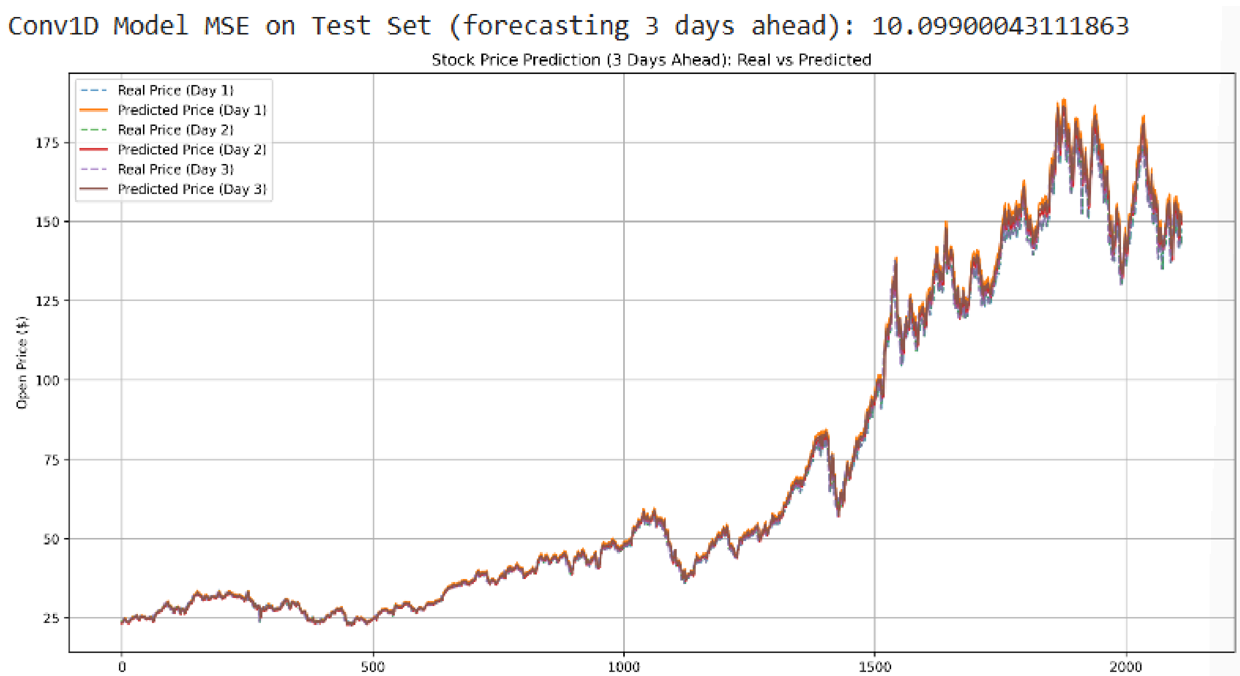

Figure 5a: The stock prediction 3 days ahead when batch size is 250.

In contrast, Figure 5b shows the model's performance with a reduced batch size of 64. While the overall patterns remain aligned between the predicted and real prices, the model exhibits a slight increase in prediction error, with the MSE rising to 30.462. This suggests that the larger batch size of 250 offers better convergence during training, leading to superior predictive performance for multi-step forecasting.

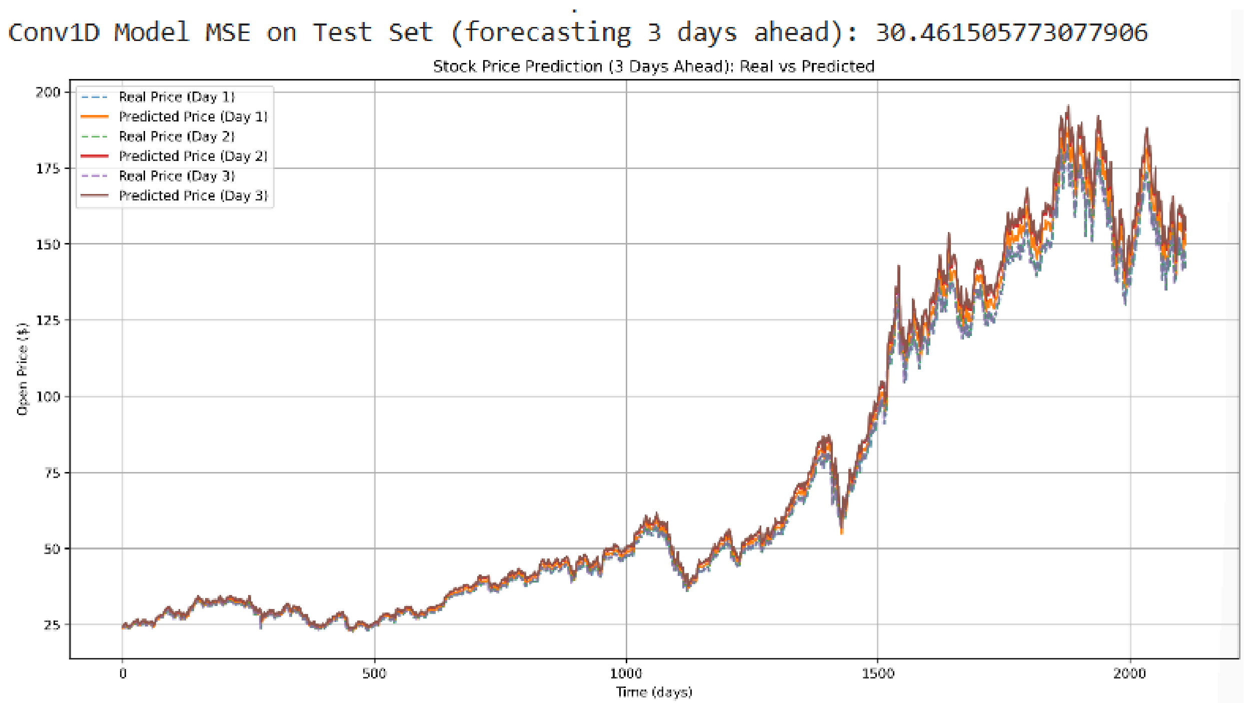

Figure 5b: The stock prediction 3 days ahead when batch size is 64.

b) When k = 7

Using the Conv1D architecture, the model was trained on sequences of 30 days of historical stock data to predict the Open price for the next 7 days, which effectively implements a longer-term forecasting horizon. Figures 6a and 6b present the model's performance under two different batch sizes: 250 and 64, respectively.

Figure 6a: The model's performance when the batch size is 250 is illustrated.

- The colored dashed lines represent the Real Prices for the next 7 days, each color corresponding to a specific day in the forecast window.
- The solid lines of matching colors represent the Predicted Prices for the same days.

From Figure 6a, the model captures the general trends of the real stock prices but exhibits noticeable deviations, especially during periods of high volatility. The Mean Squared Error (MSE) on the test set is 450.903, reflecting the challenge of accurately predicting longer-term trends due to compounding errors and the inherent uncertainty of stock price movements.

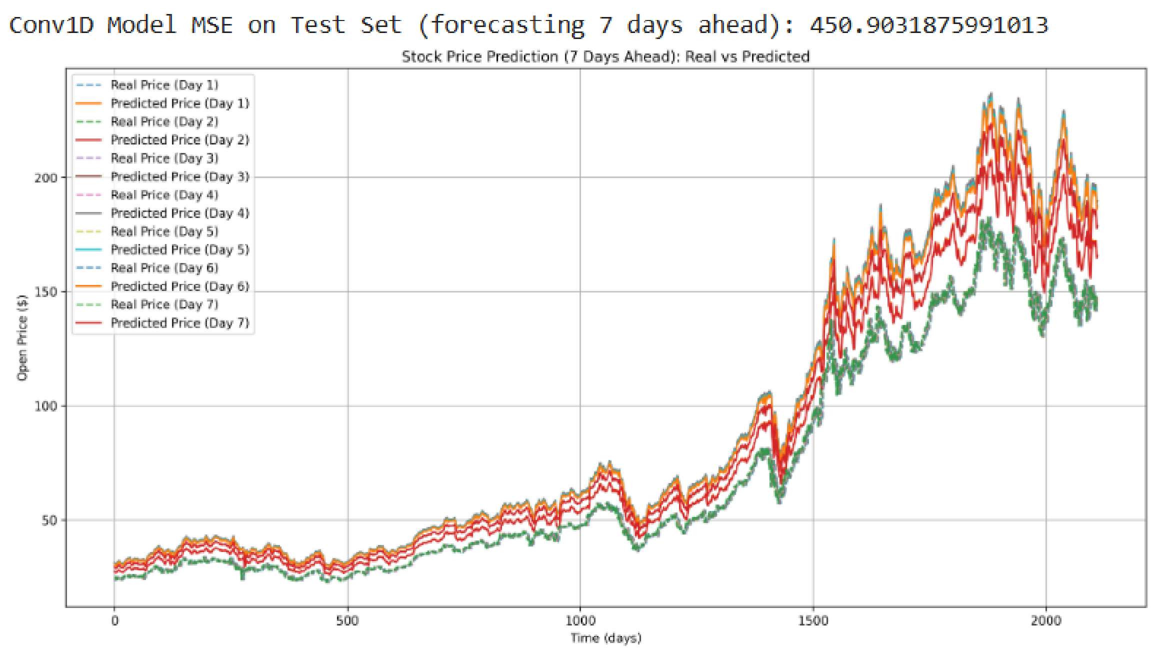

Figure 6a: The stock prediction 7 days ahead when batch size is 250.

Figure 6b: When the batch size is reduced to 64, the performance of the model improves significantly.

- The closer alignment between the Real and Predicted Prices across all 7 forecast days is clearly visible in this figure.
- The MSE is drastically reduced to 76.178, indicating a substantial enhancement in the model's prediction accuracy.

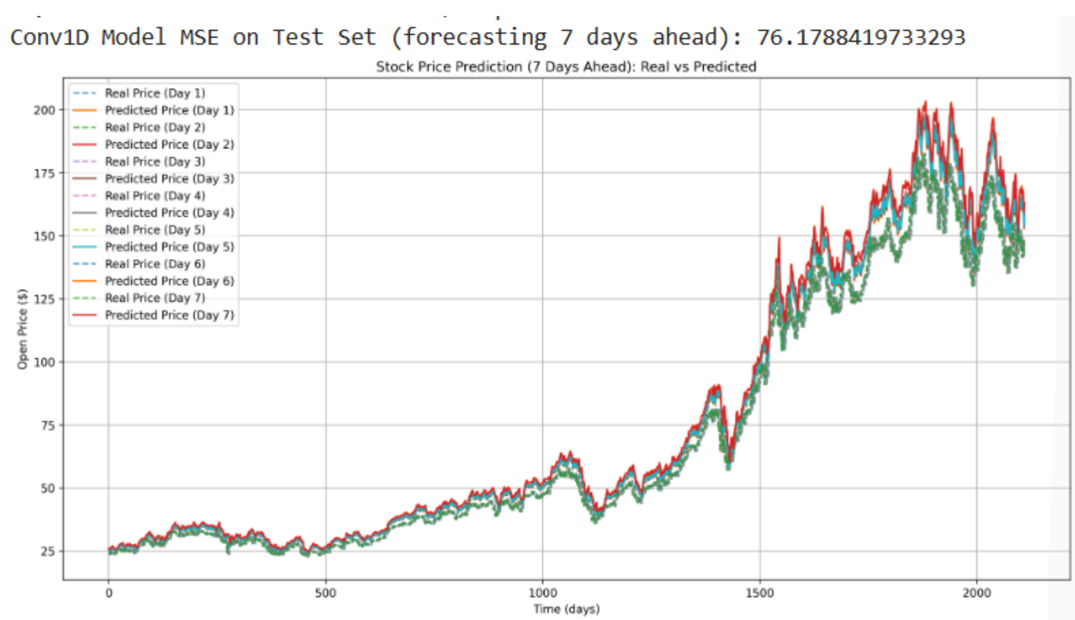

Figure 6b: The stock prediction 7 days ahead when batch size is 64.

Task 2

The objective of Task 2 is to develop a Conv1D-based model to predict the stock prices of Vietnamese company - HPG. The model uses historical stock price data, including features like open, close, high, low prices, and volume, to forecast future stock prices. The task evaluates the effectiveness of the Conv1D model in handling time-series data in the context of stock price prediction.

Task 2.1

In Task 2.1, the Conv1D model is trained to predict the stock price for a single time step (next day) based on past data. The model's performance is measured using the Mean Squared Error (MSE) as the primary evaluation metric. The model is trained using a batch size of 125. The model achieves an MSE of 95,792,001, indicating the level of error between the predicted and real stock prices. In the figure 7:

1. Orange Line: Represents the real stock prices, serving as the ground truth.
2. Blue Line: Represents the predicted stock prices by the Conv1D model.

From the figure, it is evident that the model captures the overall trend of stock price movements but struggles with high volatility and large fluctuations. This task highlights the potential of Conv1D in stock price forecasting while showcasing the challenges associated with high volatility.

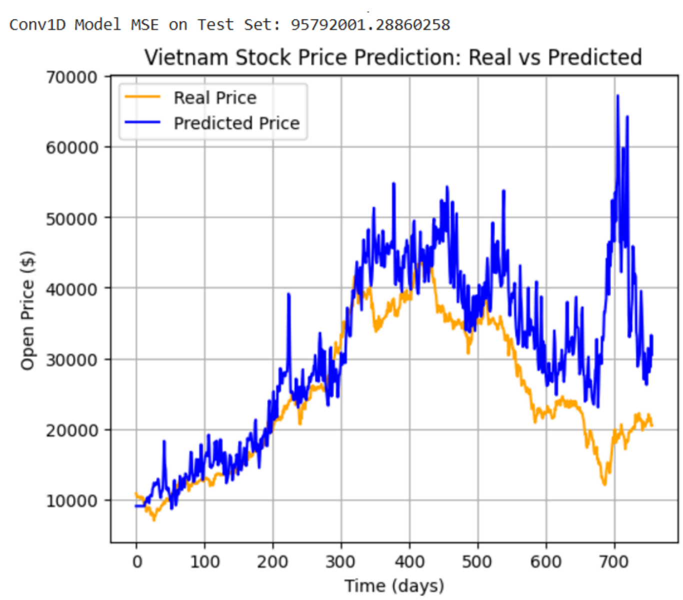

Figure 7: Vietnam Stock Price Prediction: Real vs Predicted Prices

Task 2.2

Task 2.2 explores the application of the Conv1D model for predicting Vietnam stock prices over multi-step forecasting horizons. Specifically, the objective is to predict the stock price three days (k=3) and seven days (k=7) ahead.

a) When k = 3

The model with a batch size of 64 achieved the lowest MSE of 59011686.25817944. The corresponding graph (Figure 5a) illustrates the performance of the model in predicting stock prices three days ahead. The orange line represents the actual stock prices, while the blue line represents the model's predictions. The graph shows a reasonable alignment between the predicted and actual values, with the model effectively capturing short-term price trends. The choice of a smaller batch size may have enabled the model to adapt better to the inherent variability in the dataset, leading to improved performance.

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Conv1D Model MSE on Test Set (forecasting 3 days ahead): 59011686.25817944
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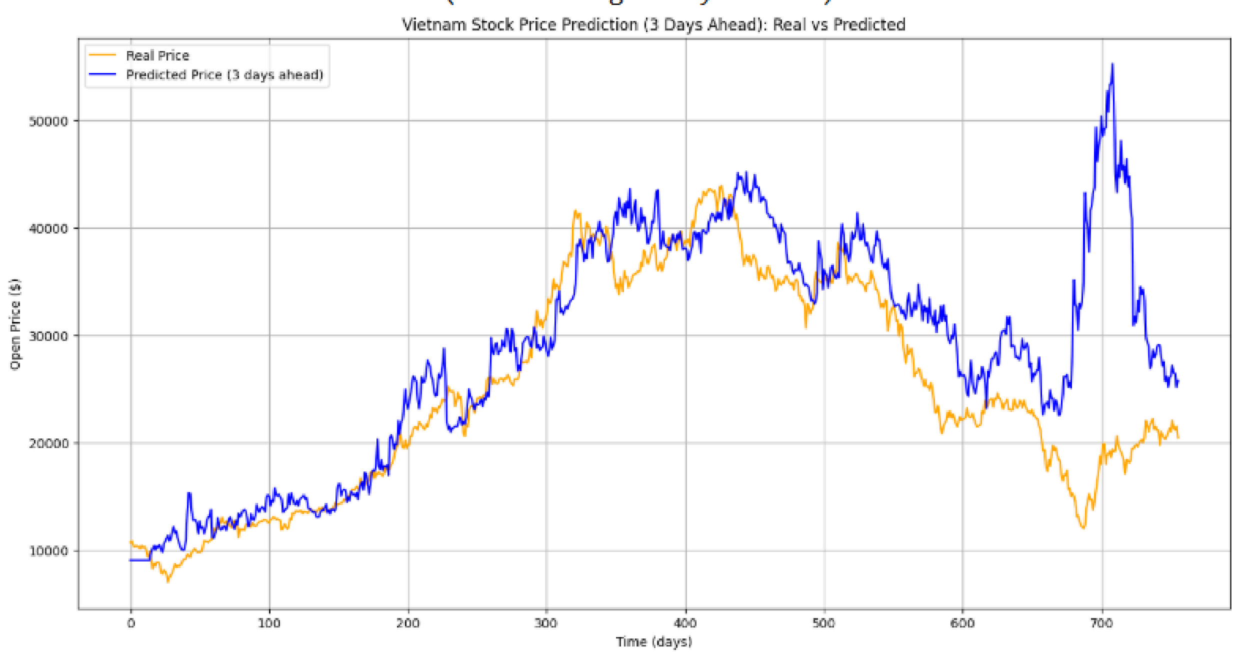

Figure 8: The stock prediction on the 3rd day when the batch size is 64.

b) When k = 7

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Conv1D Model MSE on Test Set (forecasting 7 days ahead): 84432971.90648028
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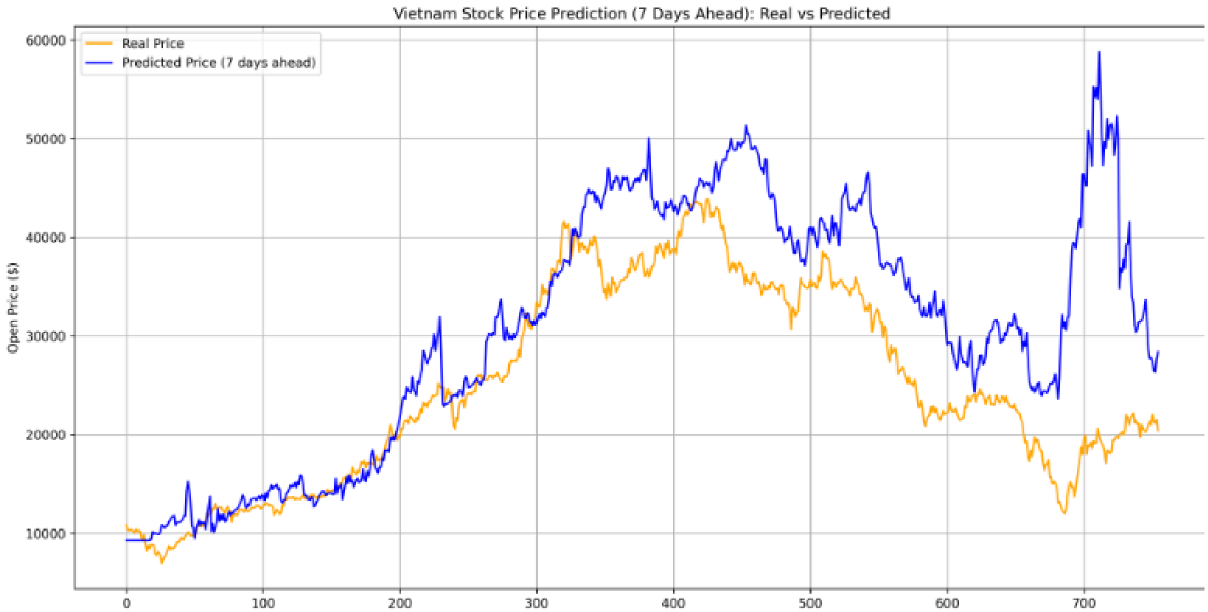

Figure 9: The stock prediction on the 7th day when the batch size is 250.

For this scenario, the best performance was observed with a batch size of 250, yielding an MSE of 84432971.90648028 (Figure 9). In the corresponding graph, the orange line depicts the real stock prices, while the blue line represents the predicted prices for seven days ahead. The model's predictions follow the general direction of the actual prices but display larger deviations compared to the three-day forecast. This is expected as the increased forecasting horizon introduces additional uncertainty and error accumulation. The selection of a larger batch size in this case likely stabilized the learning process, balancing the trade-off between generalization and convergence.

The comparison between the two configurations demonstrates that shorter forecasting windows (k = 3) result in better predictive accuracy due to reduced uncertainty in future price movements. Additionally, the optimization of batch size plays a crucial role in minimizing the MSE for each forecasting horizon, as evidenced by the lower errors achieved with specific batch sizes for both cases.

Task 2.3

For Task 2.3, we explore the performance of the Conv1D model in forecasting stock prices for multiple consecutive days. Specifically, we investigate two scenarios: forecasting for three consecutive days (K = 3) and seven consecutive days (K = 7). For each scenario, we tuned the batch size to minimize the Mean Squared Error (MSE) on the test set.

a) When k = 3

The first case focuses on forecasting stock prices for the next three consecutive days. The model achieves a test set MSE of 6,128,9208.8819. The plot in Figure 10 demonstrates the real and predicted prices for the three consecutive days. As seen in the graph, the predicted prices for Days 1, 2, and 3 closely follow the real prices, reflecting the model's effectiveness in capturing short-term trends. However, the MSE value indicates some deviation in the predictions, especially during periods of high volatility.

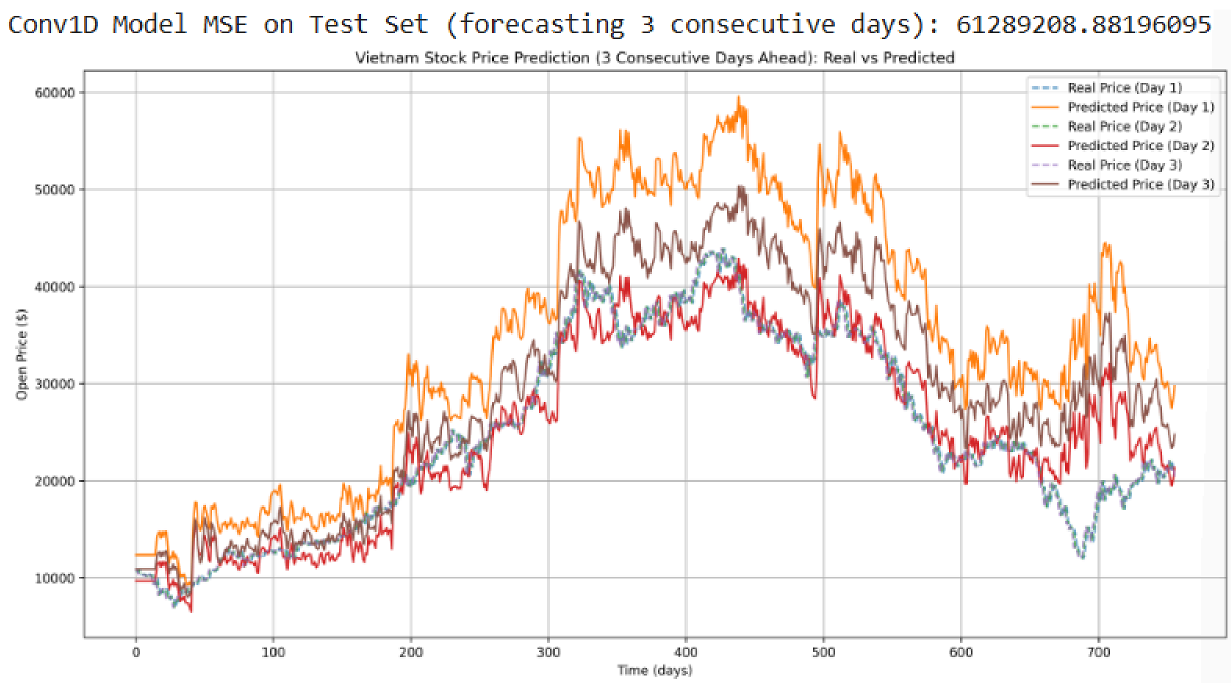

Figure 10: The stock prediction 3 days ahead when batch size is 64.

b) When k = 7

In the second case, the model predicts stock prices for the next seven consecutive days. After tuning the batch size, the model achieves a significantly lower test set MSE of 3,935,508.1645. The plot in Figure 11 displays the real and predicted prices for Days 1 through 7. While the predictions capture the overall trend, discrepancies become more pronounced with longer forecasting horizons, which is expected given the increasing uncertainty and noise over time.

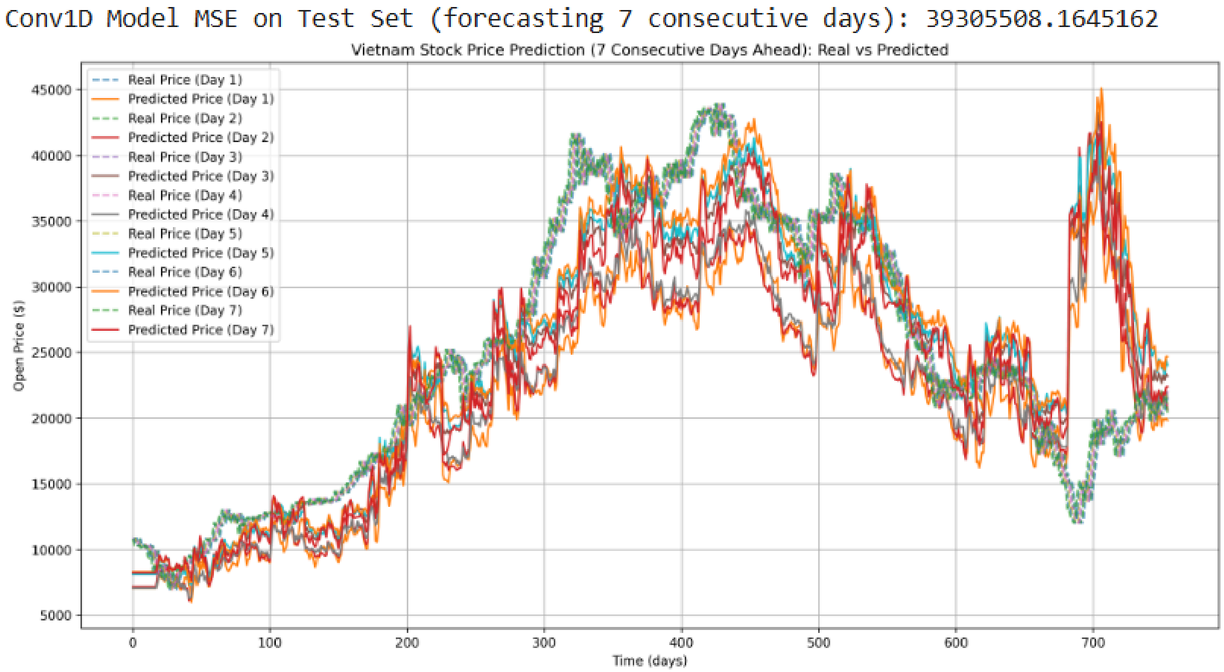

Figure 11: The stock prediction 7 days ahead when batch size is 64.

Task 3

Task 3.1

In the first attempt at Task 3.1, a binary classification model was implemented to predict buying signals, with outputs being either 0 or 1 representing "No Buy" or "Buy," respectively. However, the dataset was heavily imbalanced, with a majority of the target values being 0. As shown in Figure 12a, this imbalance caused the model to become biased, predicting mostly 0 values, leading to a failure in accurately identifying buy signals. The overdominance of 0 in the dataset resulted in poor performance and ineffective signal detection.

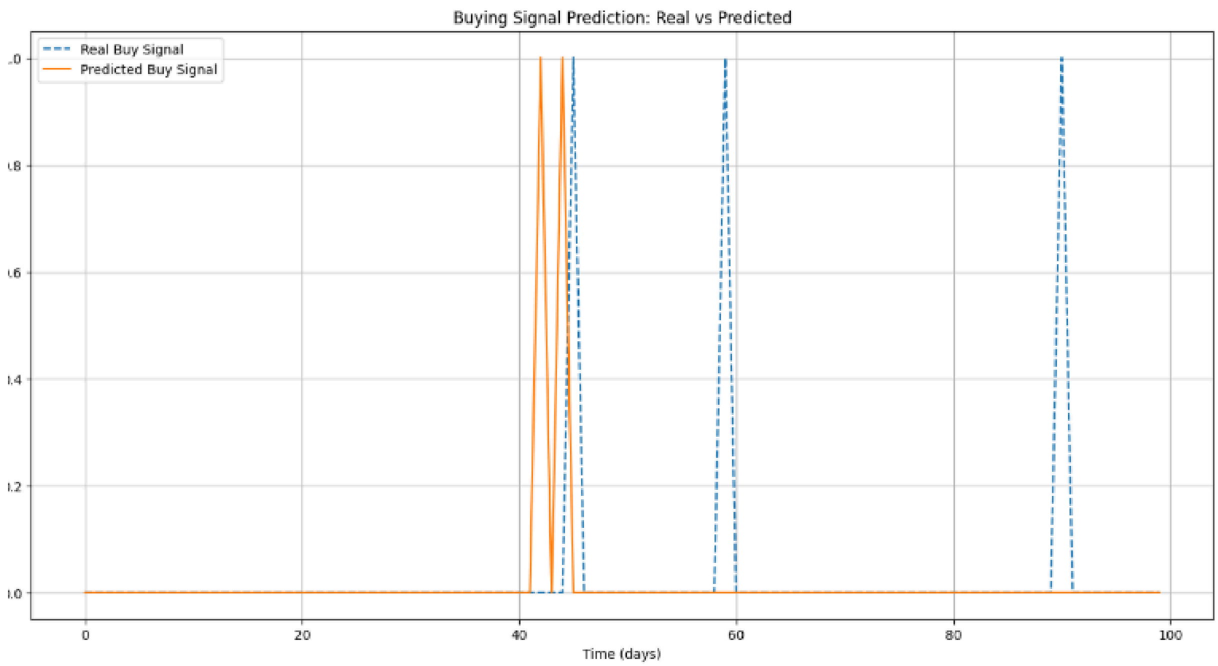

Figure 12a: Buying Signal Prediction: Real and Predicted by binary method

To address the failure of the binary classification approach, the prediction task was reformulated into a regression problem where the model outputs a probability indicating the likelihood of a buying signal. The Relative Strength Index (RSI) was introduced as a feature, and a hybrid Conv1D + LSTM architecture was utilized to process the time-series data.

Figure 12b illustrates the improved performance of this hybrid architecture, showing the predicted buy signal probabilities aligning more closely with the true probabilities. While the results were better than the initial binary classification, there were still noticeable deviations in the predicted probabilities.

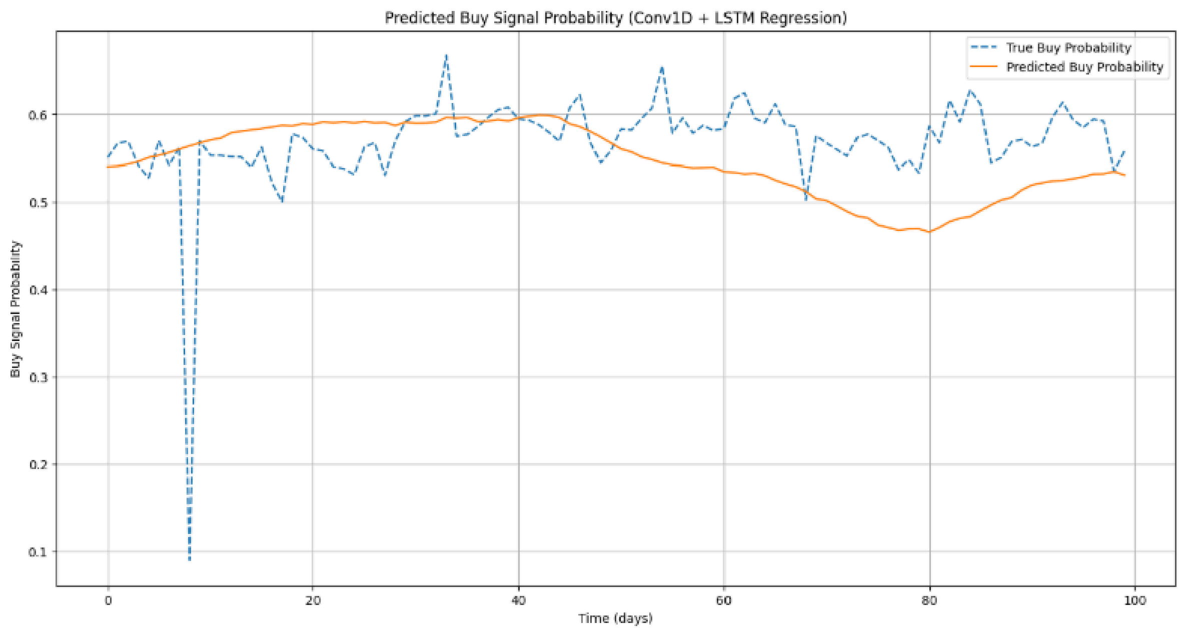

Figure 12b: Buying Signal Prediction: Real and Predicted by RSI method (Conv1D and LSTM)

Further optimization involved replacing the hybrid Conv1D + LSTM model with a pure Conv1D architecture to reduce complexity while improving prediction accuracy. This adjustment led to significant performance gains, as evidenced by the results in Figure 13c. The Conv1D model demonstrated much closer alignment between the predicted and true buy signal probabilities, with a Mean Squared Error (MSE) of 0.0011 and a Mean Absolute Error (MAE) of 0.0257. This improvement highlights the effectiveness of using a streamlined Conv1D model to capture the patterns in buying signals, leveraging features like RSI and price momentum.

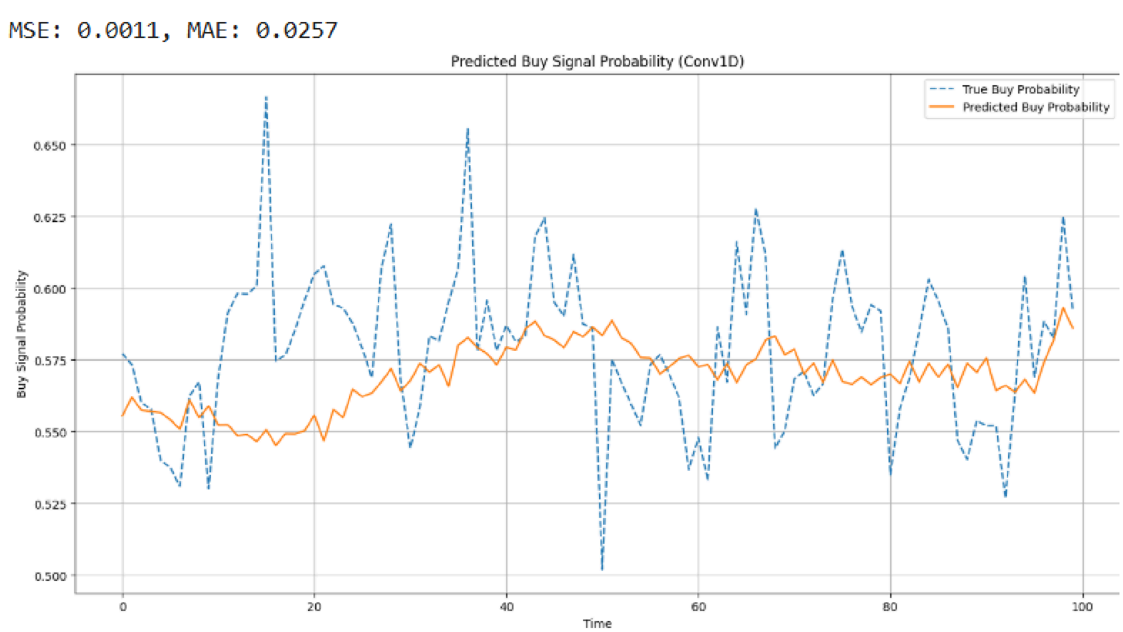

Figure 13c: Buying Signal Prediction: Real and Predicted by RSI method (Conv1D)

Task 3.2

In Task 3.2, the goal was to predict the probability of a sell signal using a regression approach. The Conv1D model was used to analyze the relationship between historical stock price patterns and the likelihood of future sell signals.

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Mean Absolute Error (MAE): 0.1723
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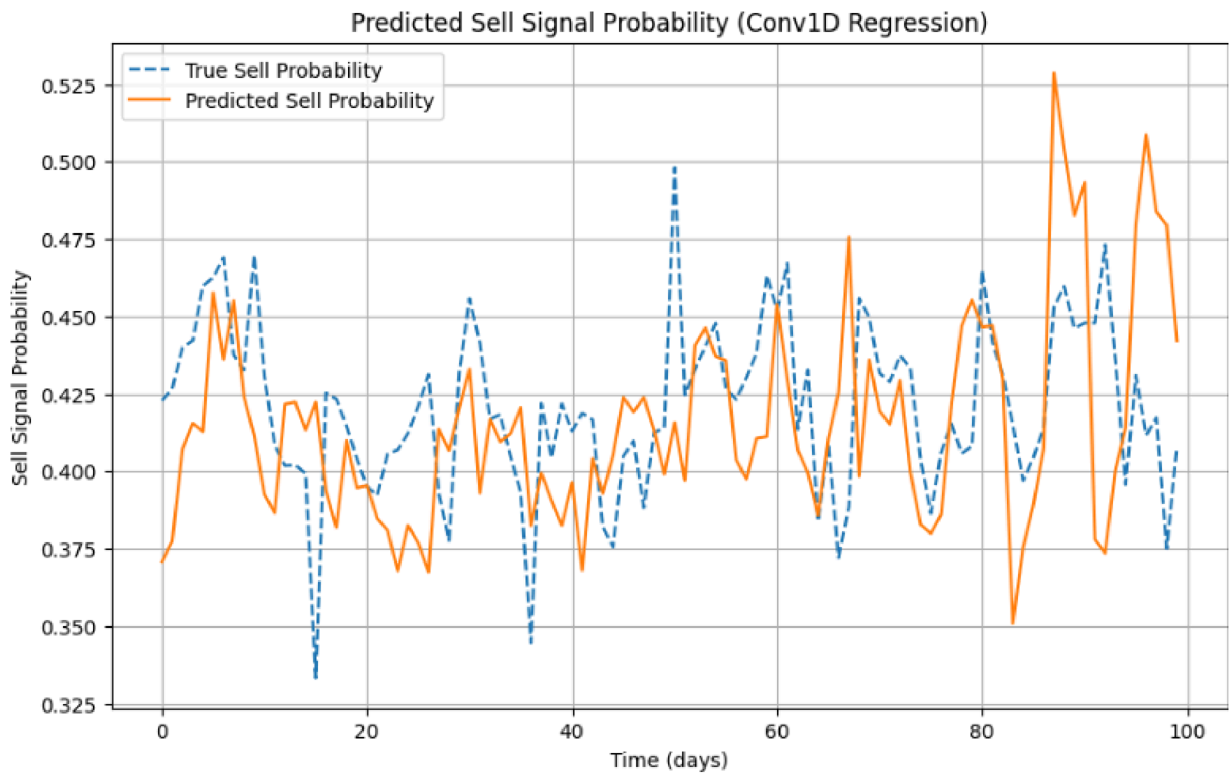

Figure 14: Predicted Selling Signal using only Conv1D

The graph in Figure 3.2 illustrates the model's predictions compared to the actual probabilities of sell signals over a specific period. The Mean Absolute Error (MAE) for this task was computed as 0.1723, indicating reasonable accuracy in the model's ability to approximate true sell probabilities.

- True Sell Probability (Dashed Line): Represents the actual probabilities of a sell signal, calculated based on future price trends.
- Predicted Sell Probability (Solid Orange Line): Represents the model's predictions for the likelihood of a sell signal.

From the visualization, the predicted sell probabilities capture some of the variations in the true sell probabilities. Despite the imperfections, the Conv1D regression approach provides a practical framework for identifying sell signals in a probabilistic manner, making it useful for informed decision-making in stock trading scenarios.

Task 4

The goal of Task 4 is to identify companies suitable for inclusion in a stock portfolio based on their profitability and risk levels. The task is divided into two sub-tasks: Task 4.1 (Portfolio Composition), which focuses on selecting profitable companies, and Task 4.2 (Risk Management), which identifies companies with high levels of risk that should be excluded from the portfolio. These analyses were based on the 7-day price change (%) of each company, which serves as a benchmark for short-term financial performance.

Task 4.1 and 4.2

```
Profitable Companies:
  Company  7-Day Price Change (%)
2     VHM                6.335938
4     VRE               15.906193
6     FPT               21.247613

Risky Companies:
  Company  7-Day Price Change (%)
0     VNM                0.000000
1     MBB              -15.958904
3     VCB               -7.497692
5     NLG              -21.292661
```

In Task 4.1, companies with positive 7-day price changes were classified as "profitable." The selected companies were VHM, VRE, and FPT, with respective price changes of 6.34%, 15.91%, and 21.25%. These results indicate that these companies outperformed others in terms of short-term growth potential, making them ideal candidates for inclusion in the portfolio.

In contrast, Task 4.2 identified companies with negative 7-day price changes as "risky." Companies such as MBB (-15.96%), VCB (-7.50%), and NLG (-21.29%) were flagged for exclusion. These results suggest that these companies have underperformed in the short term and could pose a risk to portfolio returns.

Task 4.3

We should consider practical factors like dividends and industry sector performance. Dividends reflect a company's financial strength and ability to provide steady returns to investors. Companies with strong and consistent dividend payments are generally viewed as reliable, making them favorable for portfolio inclusion.

Additionally, industry sector performance plays a key role in selecting companies. By identifying thriving sectors in the market, the portfolio can align itself with broader economic trends. For instance, if the technology sector is outperforming, a company like FPT, which is a leader in that space, can capitalize on the sector's momentum. This approach simplifies the decision-making process while ensuring the portfolio is both profitable and resilient against potential risks.

Conclusion

In summary, this exercise demonstrates the application of advanced machine learning techniques to address challenges in stock price prediction, buy-sell signal modeling, and portfolio optimization. Through improvements such as model modification, batch size modification, and calculation method modification, we can improve the accuracy of the model.